5

Responsive Design

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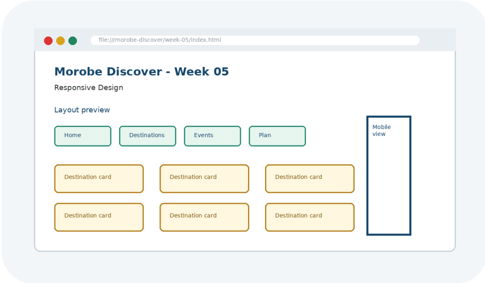

Objectives

You will have mastered the material in this chapter when you can:

- Adapt the website interface to function across various device screen sizes.
- Identify how this week builds from the previous project checkpoint.
- Read and interpret key code snippets related to the feature.
- Explain how this week prepares the next project stage.
- Explain the key concepts and terminology for this week.
- Apply the new technique to the Morobe Discover project.
- Test the weekly project increment and record evidence.

Introduction

Responsive Design introduces the next major layer of the Morobe Discover website. This chapter is written for students who are building the same project from start to final release. The topic is not treated as a separate exercise; it is a practical step in developing one working website.

A desktop-only design becomes difficult to read and use on small screens. Professional web development requires students to understand why a technique exists, what problem it solves, and how it changes the behaviour or quality of a webpage.

Project - Morobe Discover

Morobe Discover is a tourism and local discovery website. Each week adds a working layer to the project. This week the project moves from **Week 04 created structured layouts using Flexbox and Grid.** to **The same site works on mobile, tablet and desktop without separate versions.**

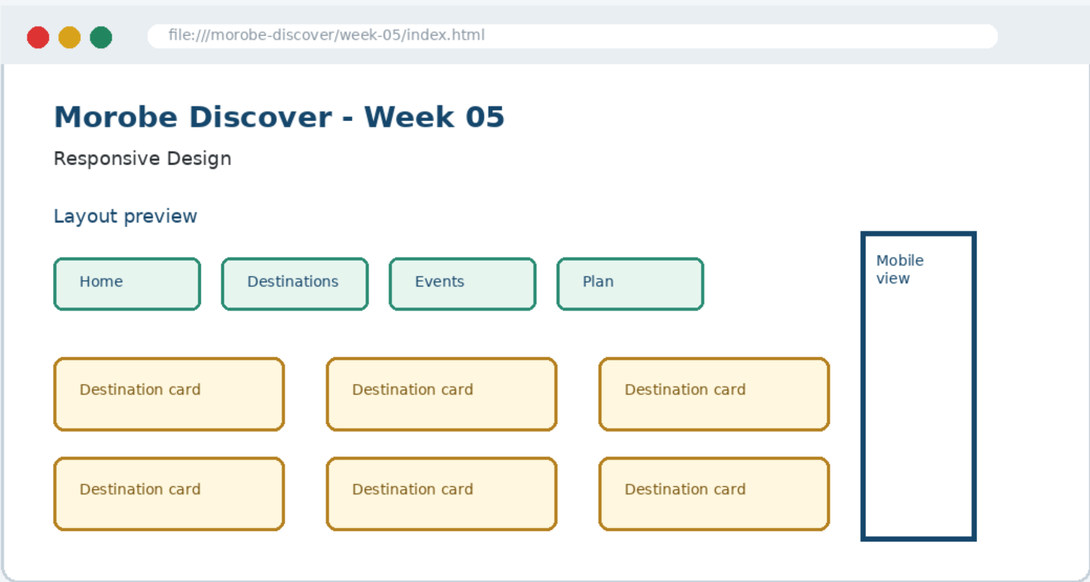

Figure 5-1. Current project focus for Week 05: Responsive Design.

By the end of the chapter, the project evidence should clearly show the development focus for this week: **Responsive rules, media queries and fluid mobile-friendly layouts**.

Before This Week and After This Week

Before Week 05	After Week 05
Week 04 created structured layouts using Flexbox and Grid.	The same site works on mobile, tablet and desktop without separate versions.
Project evidence from the previous checkpoint must be preserved.	The new feature becomes the starting point for Week 06.

Weekly Development Roadmap

1. Review current project
2. Understand the main concept
3. Inspect the interface problem
4. Study the key code
5. Apply it to the project
6. Test the result
7. Commit evidence
8. Prepare for next week

This roadmap mirrors a professional workflow. Students first inspect the current project state, then learn the concept, apply key code, test the visible result and commit the evidence.

1. Core Principle

Start with flexible layouts and add breakpoints when the layout needs help, not because of a device name.

> **Developer Tip:** A good web developer can explain not only what code was written, but why that code is appropriate for the user task and project context.

Concepts Introduced This Week

Concept	Purpose	Project use
Mobile-first CSS defines the simplest layout first	Morobe Discover application	Used in Week 05 project evidence
Media queries add layout changes at meaningful widths	Morobe Discover application	Used in Week 05 project evidence
Images and grids should be fluid	Morobe Discover application	Used in Week 05 project evidence

The concepts in this chapter are introduced in small pieces. Each piece is immediately connected to the Morobe Discover project so that students can see how the idea becomes working project evidence.

2. Explain - Illustrate - Apply

The first step is to understand the interface or coding problem. A weak implementation usually focuses on surface appearance only. A stronger implementation identifies the structural or interaction principle and applies it consistently.

Weak approach

Add code until the page appears to work. Ignore structure, reusability or testing.

Professional approach

Use the correct concept, name the affected component, test the output and commit evidence.

Thinking Question

> **Question:** How does this week improve the website for the user, not just for the developer?
> **Answer:** It makes the current project easier to use, understand, interact with, test or release.

3. Key Code Pattern A

The following code shows the smallest useful pattern for this week's implementation. It is not a complete project file. The complete project belongs in the GitHub repository under **/week-05/**.

```
img {
  max-width: 100%;
  height: auto;
}
```

Read the code line by line. Identify the selector, tag, function or command being used. Then identify the visible effect it should produce in the project.

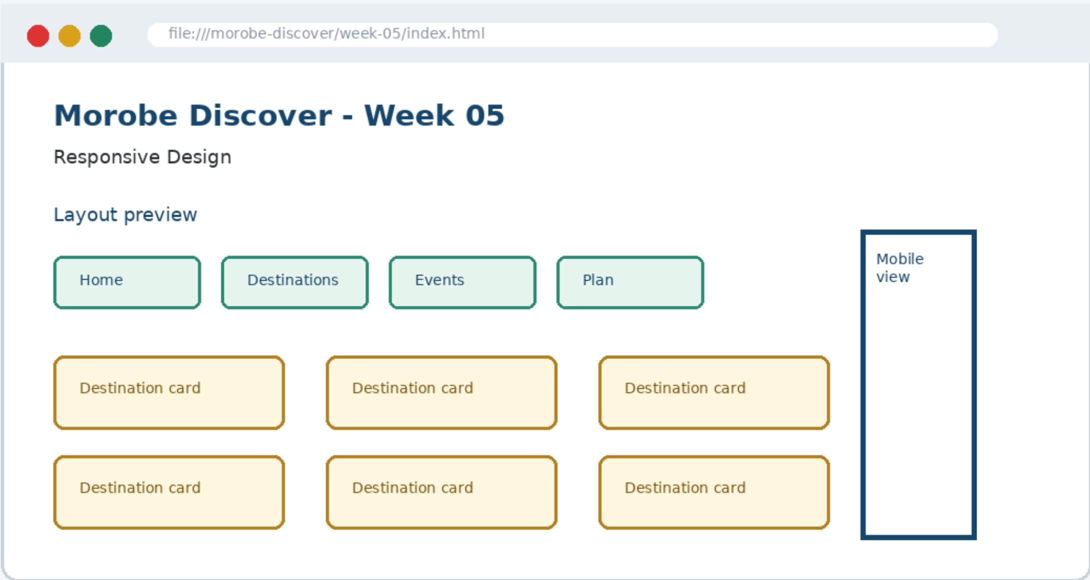

Figure 5-2. Code should be checked against visible output in the browser.

4. Key Code Pattern B

Most weeks require a second pattern that builds on the first. The second pattern usually applies the concept to a different component, state or project file.

```
@media (min-width: 768px) {
  .card-grid {
    grid-template-columns: repeat(3, 1fr);
  }
}
```

> **Good Practice:** Keep code readable, named and testable. Avoid copying code that you cannot explain during a code defence.

What to Inspect

- Which file should contain this code?
- Which part of the browser output should change?
- What error would appear if the code were placed in the wrong file?

5. Applying the Concept to Morobe Discover

Implementation should be completed as a project increment, not as an isolated demonstration. The following steps describe the development path.

To Build the Week 05 Feature

1. Stack navigation and cards on small screens
2. Add a tablet/desktop breakpoint
3. Make images and forms fluid
4. Test the main user journey at several widths
5. Record screenshots as evidence

Each step should be performed against the same Morobe Discover project folder. Do not create a separate mini-site for this week.

> **Project Rule:** Preserve earlier features. This week adds a layer to the project; it does not replace previous work.

6. Visual Result and Interface Evidence

The browser result is part of the evidence. Students should compare the interface before and after the week to confirm that the new concept has actually changed the site in the intended way.

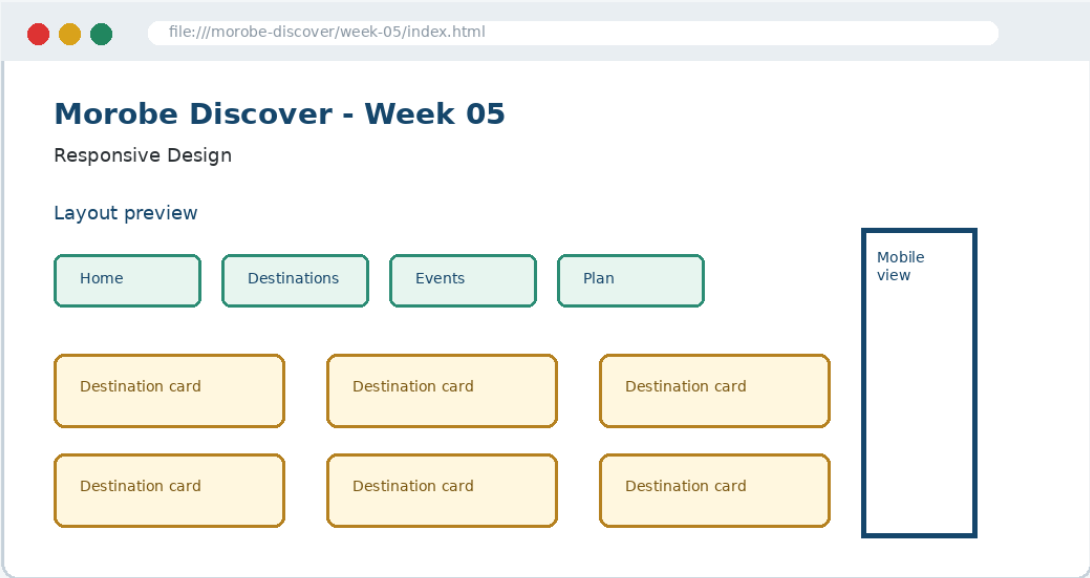

Figure 5-3. Expected Week 05 project output evidence.

Evidence item	What it proves
Screenshot	The feature is visible to the user.
Code file	The implementation exists in the correct location.
Test note	The student checked the behaviour.
Git commit	The project history records the checkpoint.

7. Common Mistake and Correction

Common Mistake: Treating this week's topic as a definition to memorise instead of a technique to apply to the project.

Incorrect approach

The student can define the concept but cannot show where it appears in the Morobe Discover files or browser output.

Correction

Open the relevant project file, locate the code pattern, refresh the browser and explain the output. A student should be able to point to the exact line of code and the exact interface region affected.

```
// Explain during code defence
File: week-05/...
Feature: Responsive Design
Evidence: browser screenshot + Git commit
```

8. Testing and Validation

Every weekly increment must be tested. Testing confirms that the feature is useful, not just present.

Check	Expected result
Use DevTools responsive mode	Expected project behaviour is visible and working.
Test 360px, 768px and desktop widths	Expected project behaviour is visible and working.
Check tap targets and line length	Expected project behaviour is visible and working.
Confirm no horizontal scrolling	Expected project behaviour is visible and working.

Incorrect result and correction: If the page appears unchanged after editing a file, confirm that you edited the correct file, linked the file properly and refreshed the browser. Then inspect the browser console where relevant.

9. GitHub and Project Evidence

The complete implementation for this week should be stored in the project repository, not copied in full into this chapter. The chapter shows key code only so that students focus on understanding.

```
Suggested repository:
https://github.com/YOUR-USERNAME/is229-morobe-discover

Weekly folder:
/week-05/

Complete project:
/complete-project/
```

Commit Message Example

```
git add .
git commit -m "Week 05 Responsive Design project increment"
```

10. Completed Weekly Outcome

At the end of Week 05, the project should demonstrate the following completed outcome:

> **Completed outcome:** Responsive rules, media queries and fluid mobile-friendly layouts

Definition of Done

Done when...	Evidence
The project includes the Week 05 feature.	Updated files in /week-05/
The feature can be explained using key concepts.	Student code explanation
The feature works in the browser.	Screenshot and test note
The work is version-controlled.	Git commit / repository history

11. Bridge to the Next Week

Week 06 adds JavaScript and DOM interaction to the responsive interface.

Week 05 output	Next use
Responsive rules, media queries and fluid mobile-friendly layouts	Week 06 adds JavaScript and DOM interaction to the responsive interface.

This bridge is important because the course uses one cumulative project. The work produced this week becomes part of the starting point for the next class.

Chapter Summary

This chapter introduced Responsive Design as a practical development step in the Morobe Discover project. You learned the principle behind the topic, studied key code patterns, applied the concept to the project, tested the result and prepared evidence for GitHub.

Apply What You Learned

1. Find the first browser width where the card layout begins to fail.
2. Identify the exact project file that should change this week.
3. Explain how the browser output proves the feature works.
4. Write one test note for the weekly feature.
5. Explain how this week connects to the next week.

> **Before the next class:** Read the next week's material and bring your current project with this week's feature completed and committed.